

# PyGameMaker — Tutorial 7: Platformer — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Platformer: Run, Jump, Collect**, 4 pages) and the matching student handout and worksheet. Students should have finished Tutorials 2 and 6 (events, collisions, solid objects, score).

## Overview

Students build a platformer in three phases: gravity, running and jumping; coins, spikes and a flag; and a score and lives display. New ideas: **gravity**, **jumping** with a vertical-speed impulse, **hazards**, and lives that survive a room restart.

Gravity is a good place to ask "what does the computer do every step?": add 0.5 to the vertical speed, then move. A jump is just a big upward speed that gravity slowly cancels.

## Suggested Timing (60 minutes)

The tutorial says 25-30 minutes for a confident student; level building adds time.

| Segment                            | Time      | What happens                                            |
|------------------------------------|-----------|---------------------------------------------------------|
| Recap and predict                  | 5 min     | Ask what forces act on a jumping character              |
| Phase 1: gravity, running, jumping | 15-20 min | Pages 1-2; the level layout takes time                  |
| Phase 2: coins, spikes, flag       | 15 min    | Page 3                                                  |
| Phase 3: controller and lives      | 10 min    | Page 4                                                  |
| Play and tune                      | 5 min     | Change gravity and jump speed; play each other's levels |
| Worksheet / exit ticket            | 5 min     | Worksheet parts A-C                                     |

For a 45-minute slot, do Phases 1-2 and leave Phase 3 for homework.

## Step-by-Step Walkthrough and Common Issues

### Phase 1: gravity, running, jumping

- Gravity is set once in the player's **Create** event: direction **270** (down), strength **0.5**. Angles start at 0 (right) and go counter-clockwise; 90 is up.
- No key** must be *\*Set horizontal speed to 0\**, **not** *\*Stop movement\**. Stop Movement also wipes the vertical speed that gravity builds, so the player hangs in the air (in the reference project it drifts about 15 pixels in half a second instead of falling more than 100). This is the most common cause of "the player does not fall".
- The jump is on **Key press** (once per press), not *\*Keyboard (held)\**.
- The ground needs to be **Solid**, and the player needs **Collision with obj\_ground** with *\*Stop movement\**; otherwise it falls through.
- Keep the space above the player free** when placing them. A platform directly above the starting position blocks the jump and looks like "the jump does not work". The jump in the reference project is about 95 pixels (three tiles) high.

- Players must be placed standing on the ground (or above it). Snap to Grid 32×32 makes that easy.

### Phase 2: coins, hazards and flag

- All the interactions are collision events **on the player**; the coin, spike and flag objects need no events.
- The spike uses \*Restart room\*. In Phase 2 nothing is remembered between restarts; the coins come back.
- The flag shows a message; there is no next level yet.

### Phase 3: game controller and lives

- The controller must be placed in the room.
- **Set the score and lives in Game Start, not Create.** A Create event runs again every time the room restarts, and a spike restarts the room, so with Create the lives were reset to 3 after every spike and could never run out. (The tutorial used Create until this was corrected; if a student's screen shows "Create", it is an old copy of the instructions.) Game Start runs once, at the very start of the game.
- With Game Start, the score is also kept across restarts, so a student can collect the same coins again after each death. Ask students whether that is fair; resetting the score in the spike event is a good change.
- Nothing special happens when the lives reach 0 (no Game Over): that is a challenge at the end of the tutorial.

**Reference projects.** Download the checkpoint projects from the wiki (07\_platformer\_checkpoints.zip): one project for the end of each phase, with a small level. Give a stuck student the previous phase.

## Vocabulary Introduced

| Term           | What it means here                                                 |
|----------------|--------------------------------------------------------------------|
| Gravity        | Adds to the vertical speed every step, in the given direction      |
| Vertical speed | Up is negative, down is positive                                   |
| Impulse        | A single change of speed, like the jump                            |
| Hazard         | An object that harms the player (spike)                            |
| Game Start     | An event that runs once when the game begins, not on room restarts |

## Discussion Questions

- Why is the jump speed negative?
- What happens to the jump if we double the gravity? Half the jump speed?
- Why does the player need a collision event with the ground as well as the ground being Solid?
- Why can't the lives be set in Create?

## Differentiation

- **Support:** give the Phase 1 checkpoint and let students build the level on top; keep the level small.
- **Extension:** moving platforms; patrolling enemies; a double jump; Game Over at 0 lives; a second room from the flag.

## Worksheet Answer Key

**Part A:** 1-C, 2-D, 3-A, 4-B. **Part B:** 1. Create (Set gravity). 2. Key press: Up Arrow. 3. Collision with obj\_coin (on the player). 4. Game Start (on the controller). 5. Draw (on the controller). **Part C:**

- 1 About 20 steps ( $10 \div 0.5 = 20$ ).
- 2 \*Stop movement\* in **No key**; it should be \*Set horizontal speed to 0\*.
- 3 The room restarts and the Create event runs again, so the lives are set back to 3 after every spike and never run out. It has to be Game Start.

**Parts D and E:** completion and reflection.

## Rubric: Platformer

| Level            | What the game shows                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 - Complete     | Gravity, running and one jump per press work; platforms are solid; coins score and disappear; spikes cost a life that stays lost; flag shows a message; score and lives are displayed |
| 3 - Working      | Movement, jumping, coins and spikes work; lives or display is missing                                                                                                                 |
| 2 - Partly there | The player falls, runs and jumps, but coins or hazards do not work                                                                                                                    |
| 1 - Started      | The player and ground exist, but the player falls through or cannot move                                                                                                              |

## End-of-Session Checklist

- ☐ Every student has a saved project (Ctrl+S)
- ☐ Every student reached at least Phase 1 (run and jump)
- ☐ Note who needs the checkpoint project next time